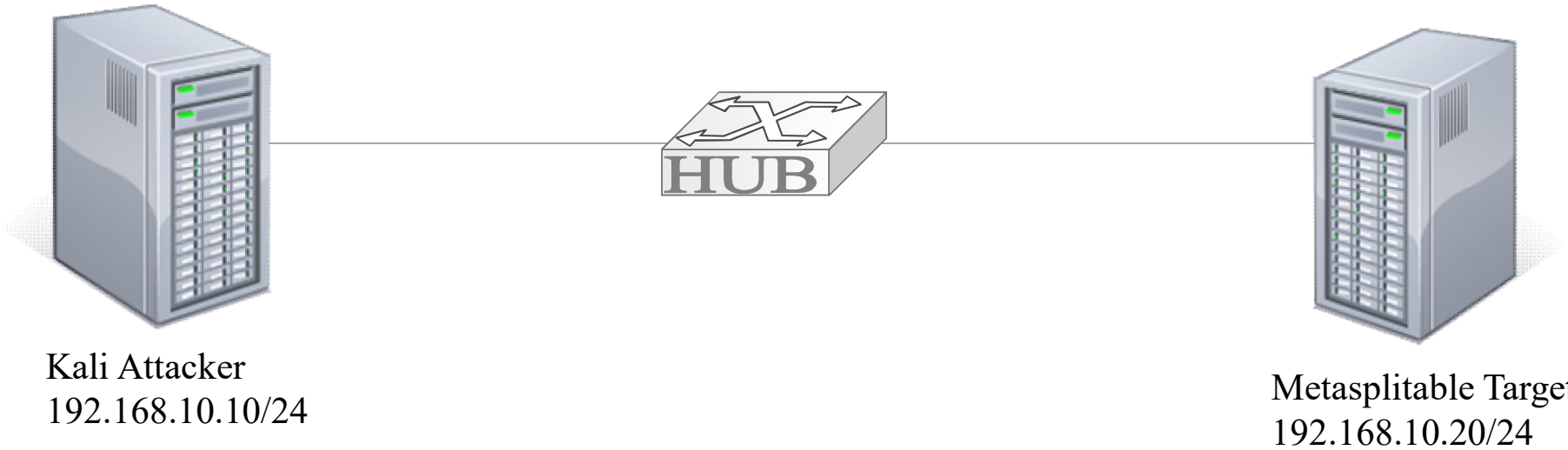
실습환경


HUB
Kali Attacker
192.168.10.10/24
Metasplitable Target
192.168.10.20/24

설치 소프트웨어

① 운영체제

•Windows 10 (64bit)

② 다운로드 파일

- VMware-workstation-full-16.1.0-17198959.exe
- kali-linux-2021.1-vmware-amd64.7z
- metasploitable-linux-2.0.0.zip

다운로드 파일 ❶

- 아래 사이트 링크 후 해당 파일 다운로드

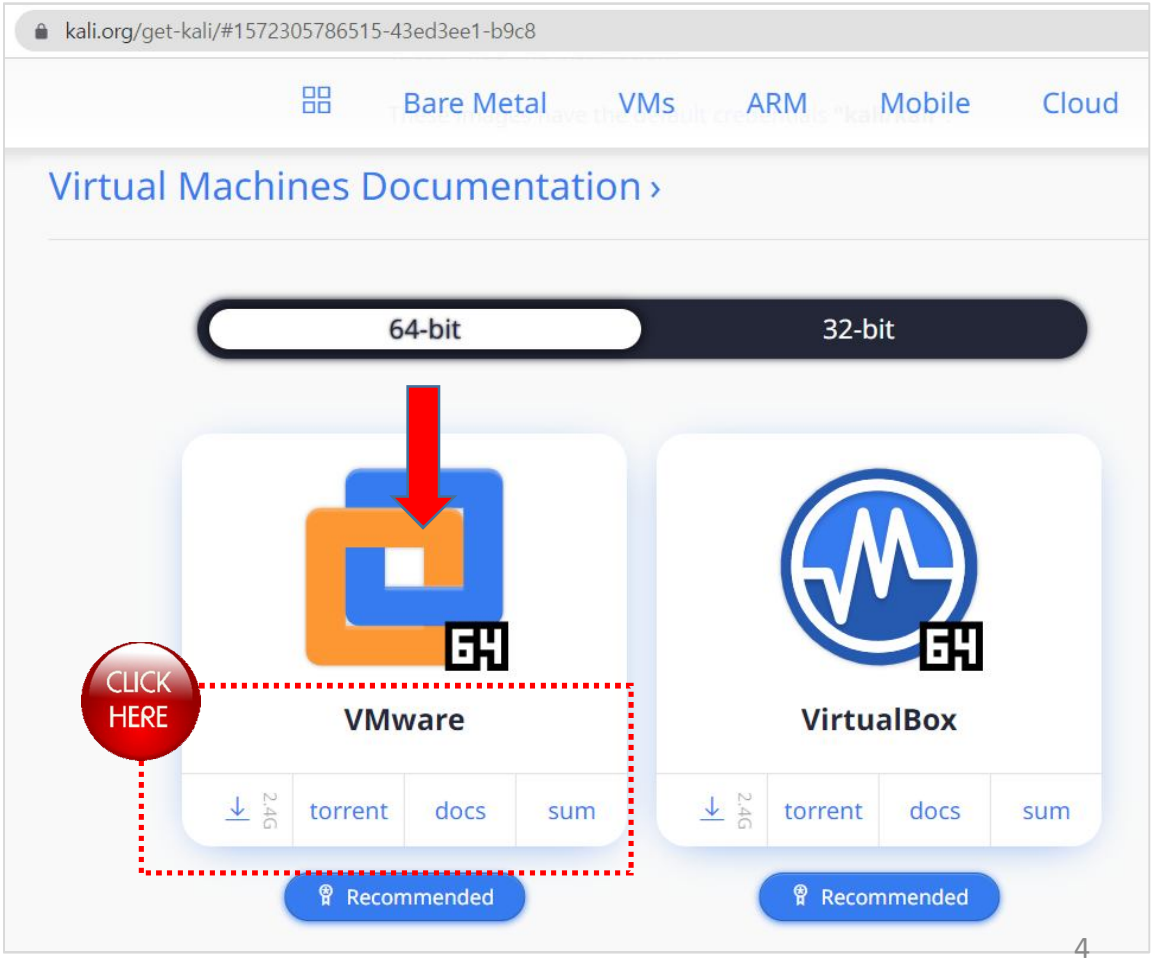
https://www.vmware.com/products/workstation-pro/workstation-pro-evaluation.html

다운로드 파일 ❷

https://www.kali.org/get-kali/#1572305786515-43ed3ee1-b9c8

➔ **kali-linux-2022.3-vmware-amd64.7z**

해당 메뉴는 Scroll bar로 아래로 내려오면 있음

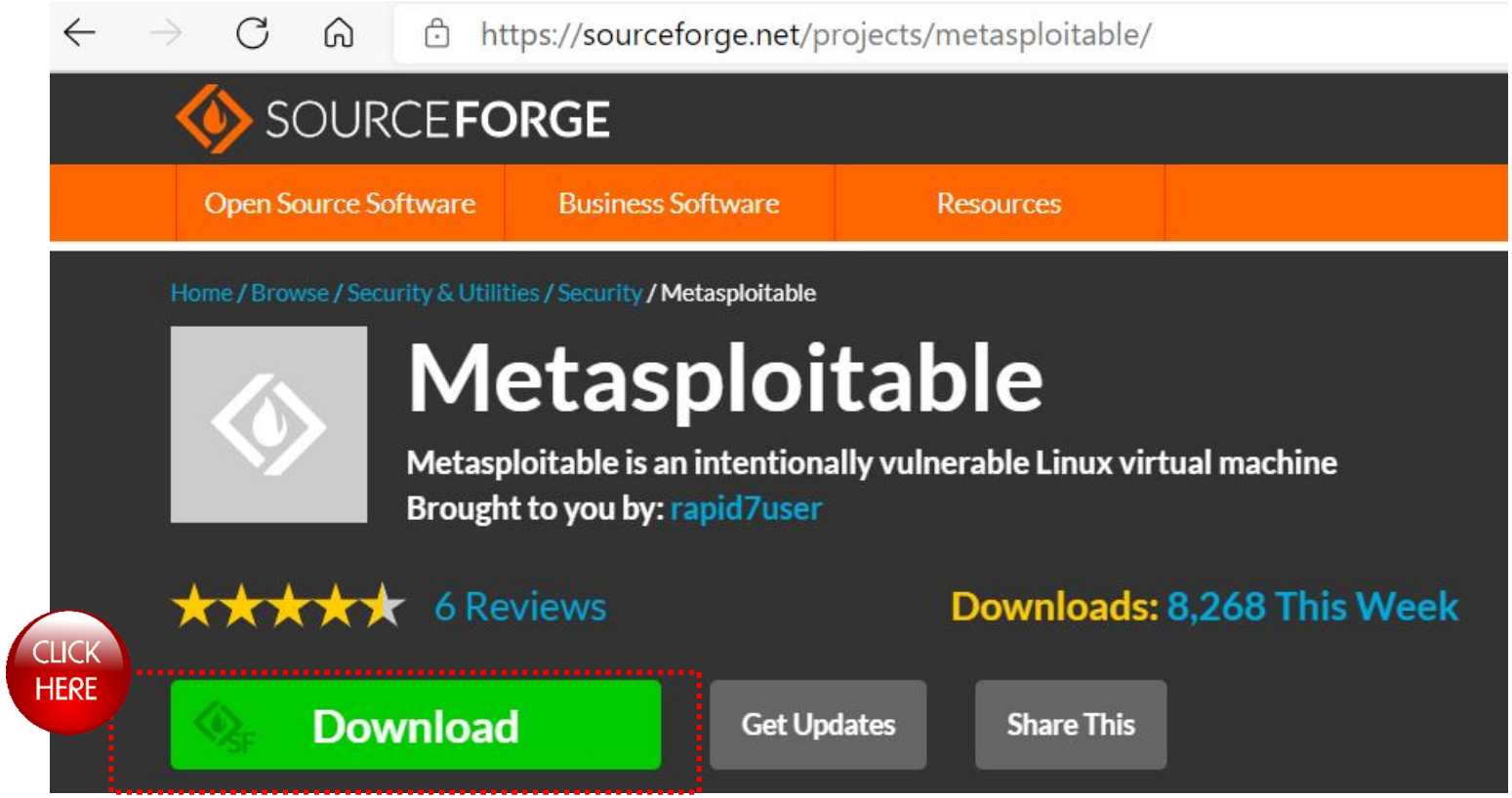


다운로드 파일 ❸

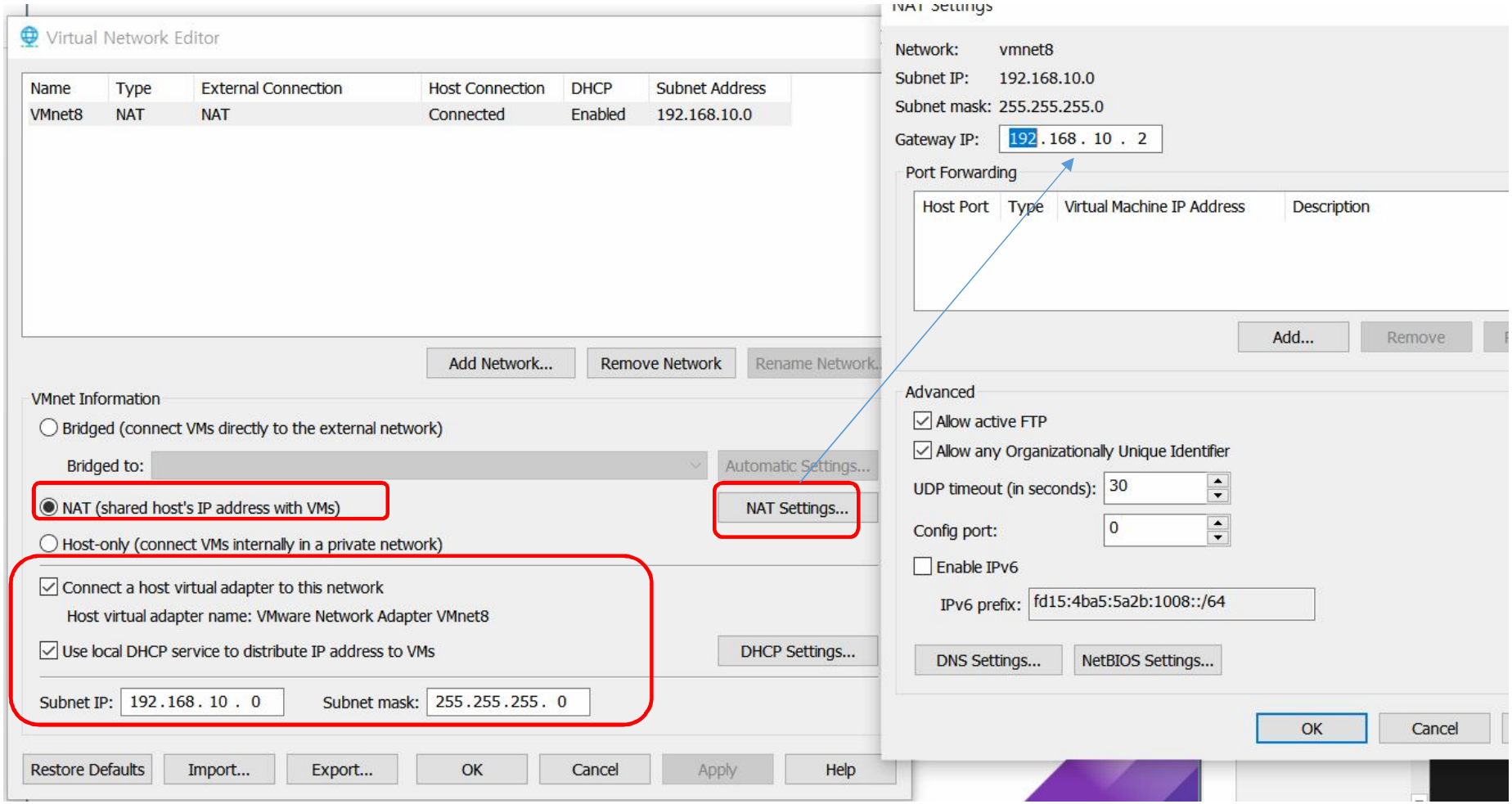
- 아래 사이트 링크 후 해당 파일 다운로드

  https://sourceforge.net/projects/metasploitable

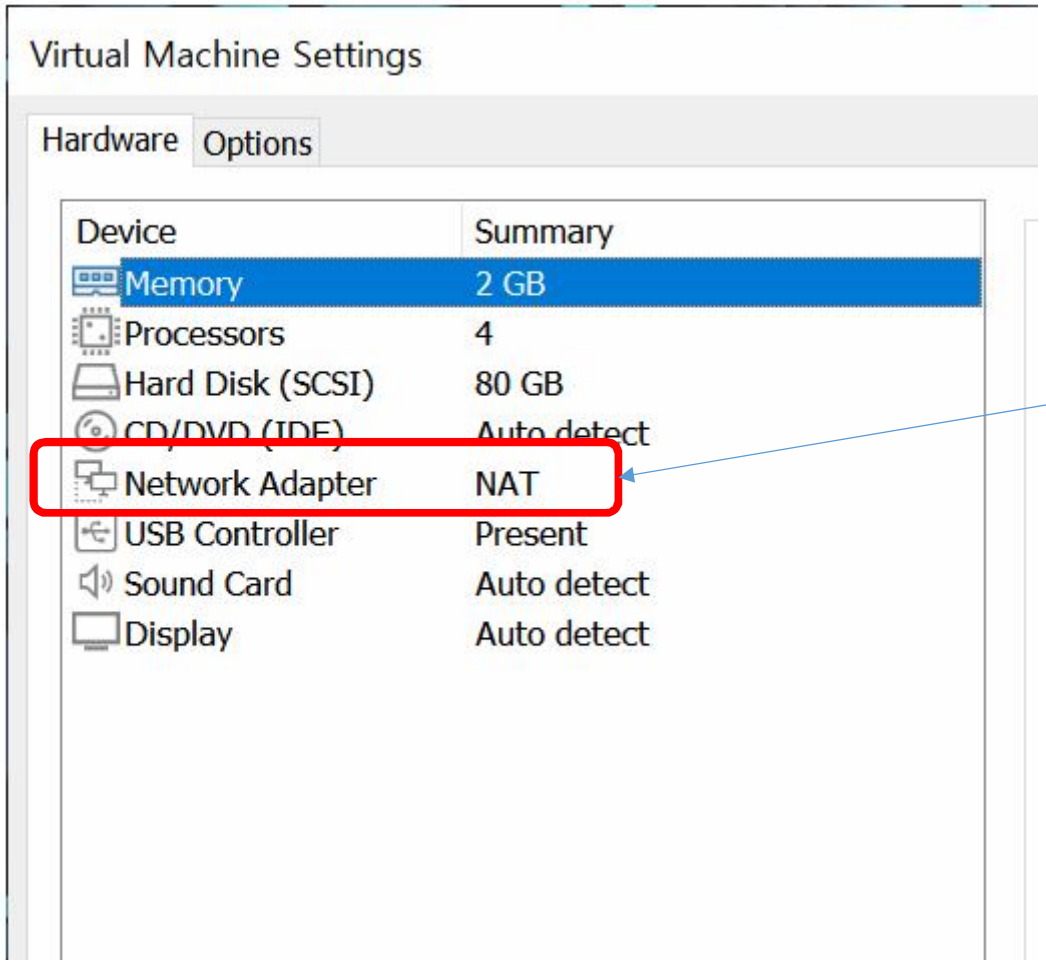
  ➔ **metasploitable-linux-2.0.0.zip**



VMWare > Edit > Virtual Network Editor



Kali/Meta - 네트워크 설정 확인



NAT 설정 확인

Kali - 네트워크 설정 확인

ID/Password : kali/kali

ifconfig

IP address는 다를수 있음

```
kali@kali: ~
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.10.128  netmask 255.255.255.0  broadcast 192.168.10.255
        inet6 fe80::20c:29ff:fe26:d083  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:26:d0:83  txqueuelen 1000  (Ethernet)
        RX packets 62079  bytes 74597762 (71.1 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 17987  bytes 2278826 (2.1 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 625  bytes 205598 (200.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 625  bytes 205598 (200.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Meta - 네트워크 설정 확인

ID/Password : msfadmin/msfadmin

ifconfig

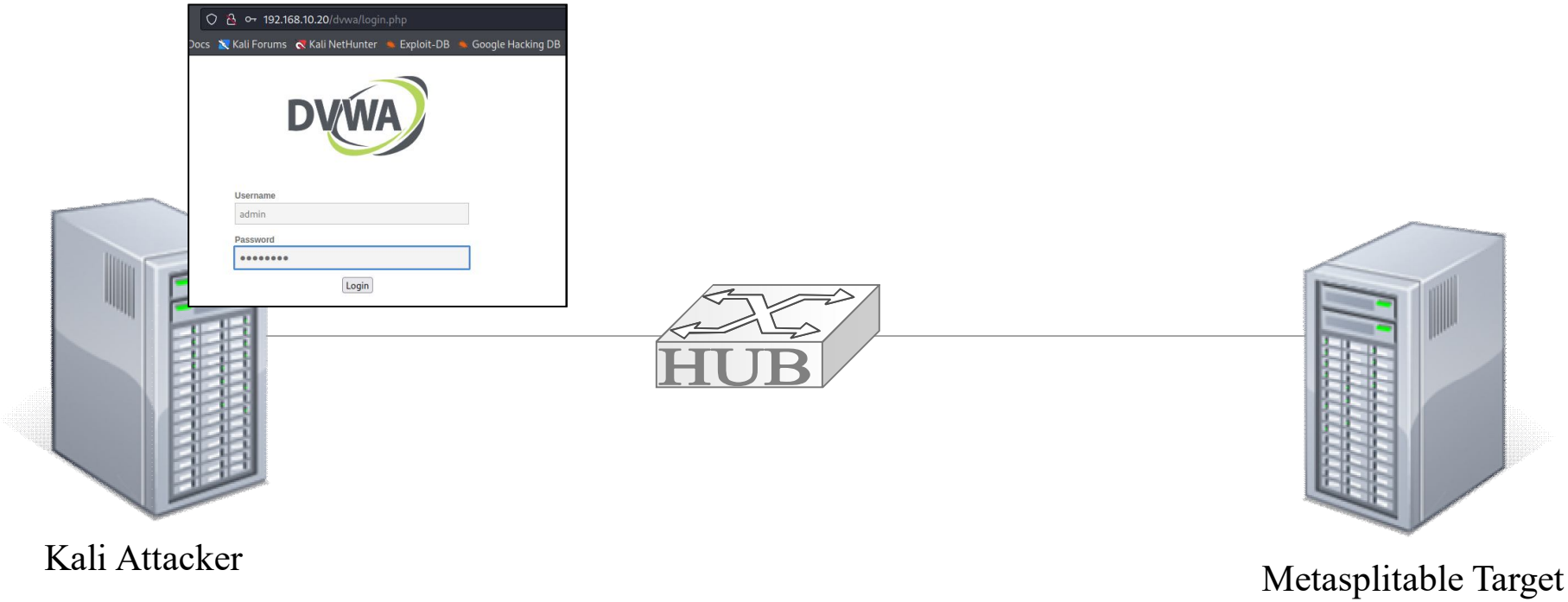
IP address는 다를수 있음

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:67:d2:b9
          inet addr:192.168.10.129  Bcast:192.168.10.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe67:d2b9/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1416 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1546 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:207792 (202.9 KB)  TX bytes:937716 (915.7 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:3389 errors:0 dropped:0 overruns:0 frame:0
          TX packets:3389 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1670221 (1.5 MB)  TX bytes:1670221 (1.5 MB)

msfadmin@metasploitable:~$
```

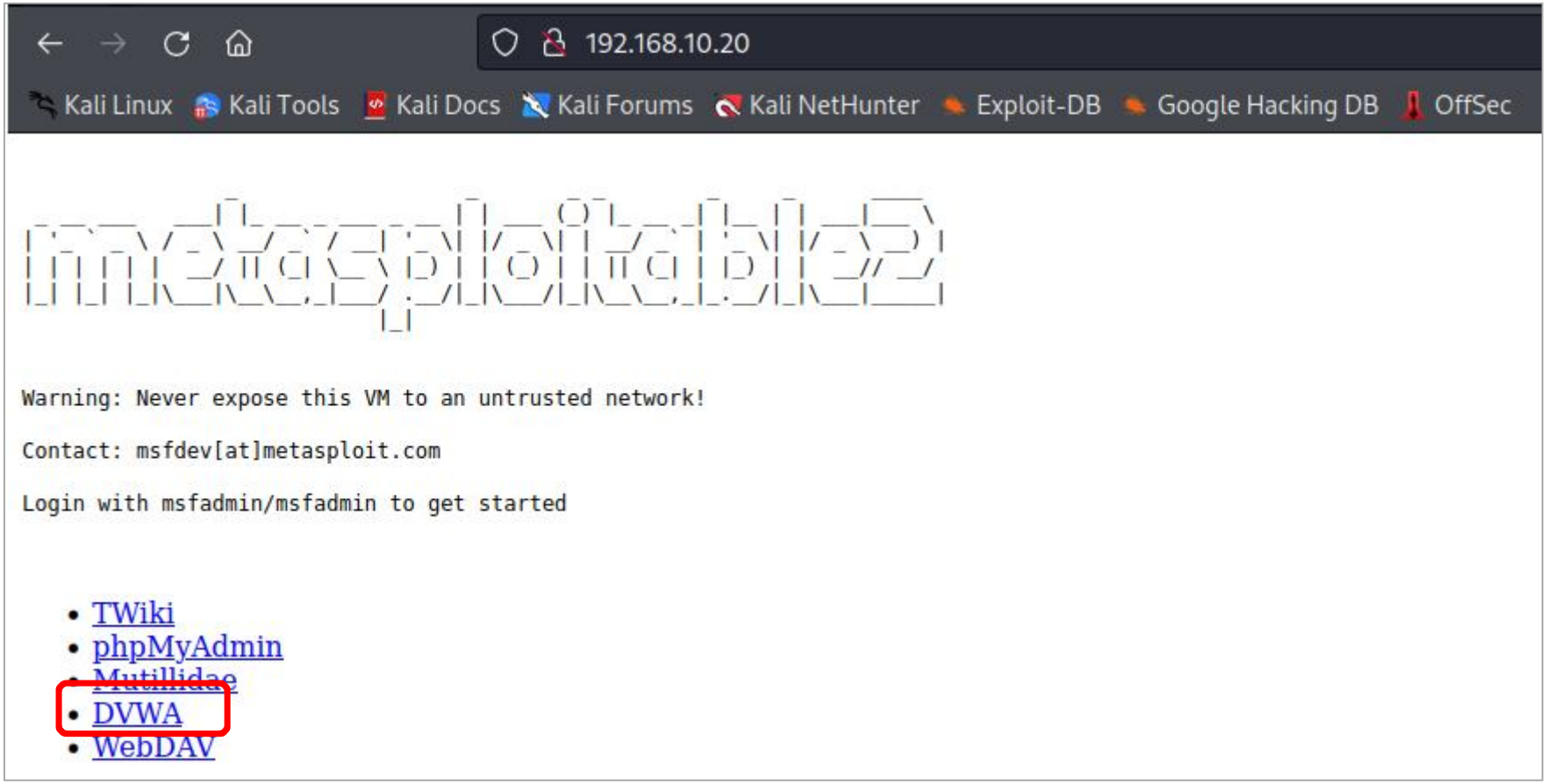
Kali ➔ Meta로 ping test

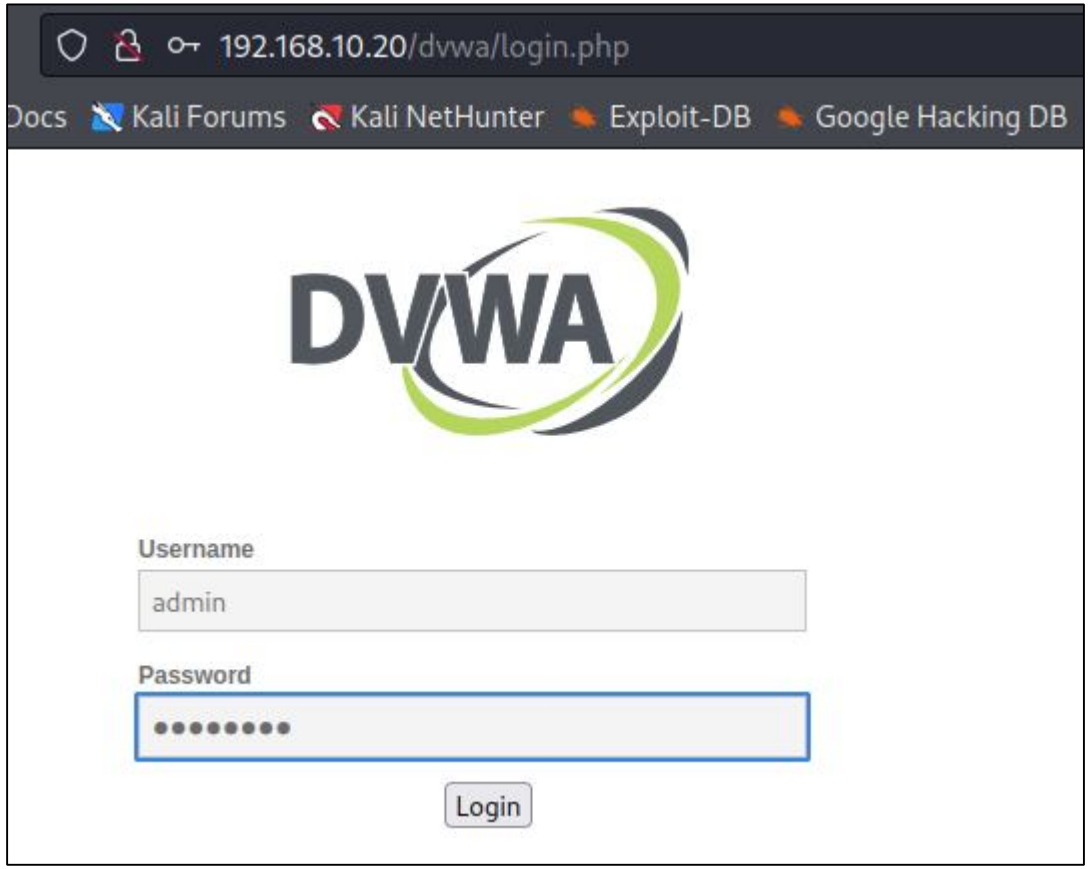
ping 192.168.10.129

```
kali@kali: ~
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ ping 192.168.10.129
PING 192.168.10.129 (192.168.10.129) 56(84) bytes of data.
64 bytes from 192.168.10.129: icmp_seq=1 ttl=64 time=7.81 ms
64 bytes from 192.168.10.129: icmp_seq=2 ttl=64 time=0.941 ms
64 bytes from 192.168.10.129: icmp_seq=3 ttl=64 time=1.03 ms
64 bytes from 192.168.10.129: icmp_seq=4 ttl=64 time=0.962 ms
^C
--- 192.168.10.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.941/2.684/7.807/2.957 ms

┌──(kali㉿kali)-[~]
└─$
```


192.168.10.20/dvwa/login.php
Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
DVWA
Username
admin
Password
Login
HUB
Kali Attacker
Metasplitable Target

[Kali] http://192.168.10.20



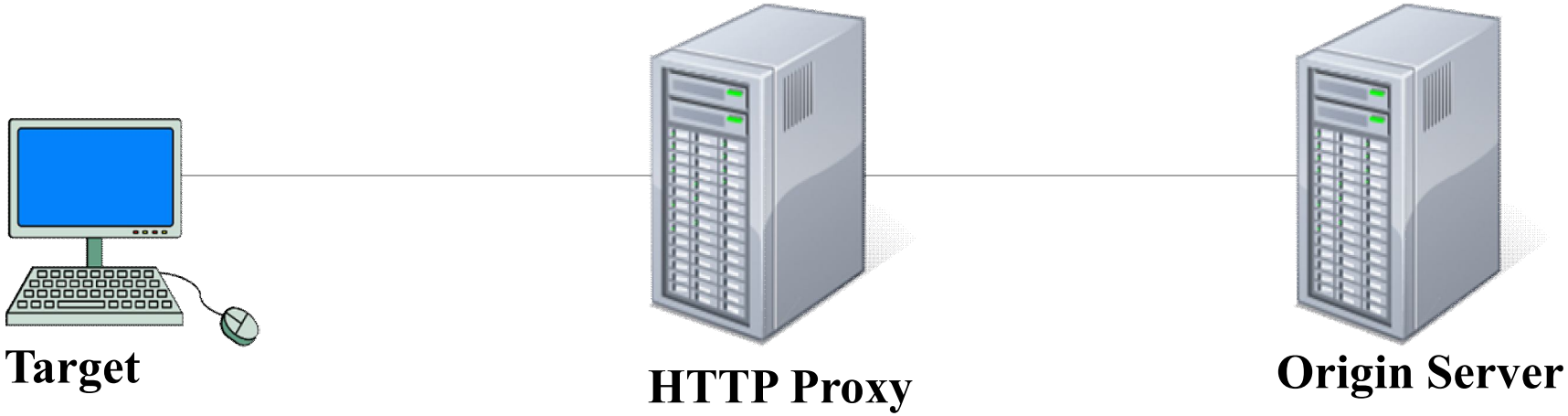


Username : admin

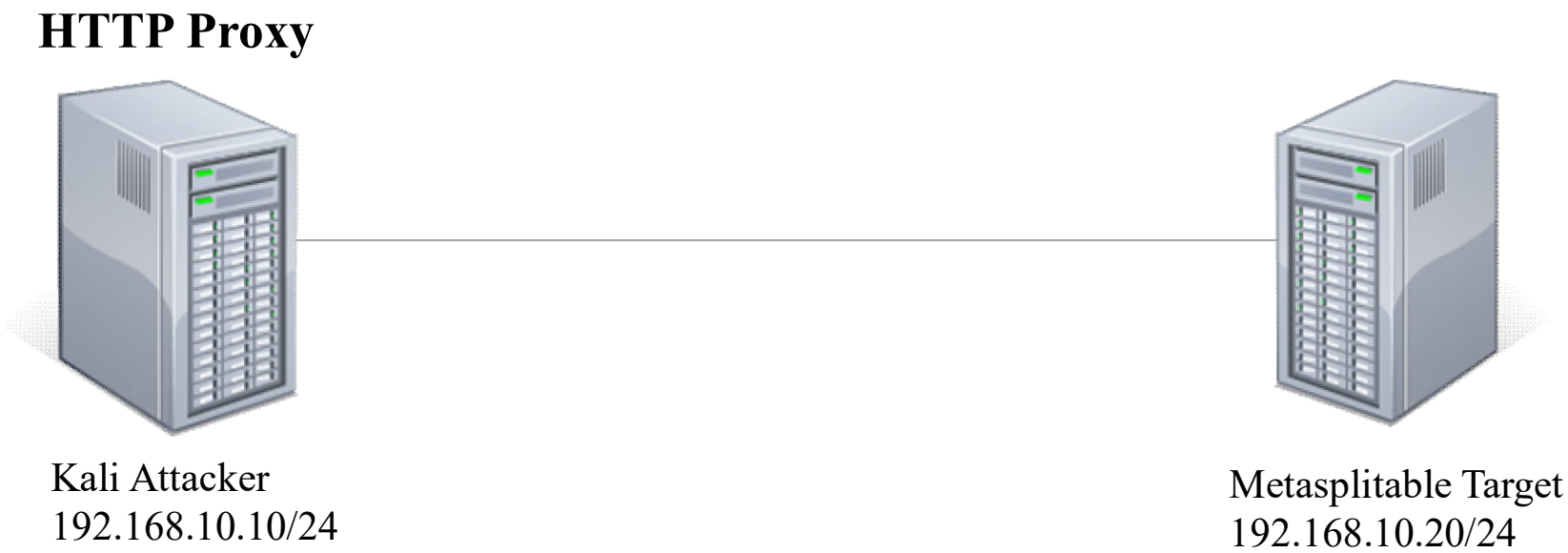
Password : password

Home
Instructions
Setup
Brute Force
Command Execution
CSRF
File Inclusion
SQL Injection
SQL Injection (Blind)
Upload
XSS reflected
XSS stored
DVWA Security
PHP Info
About
Logout
DVWA Security
Script Security
Security Level is currently high.
You can set the security level to low, medium or high.
The security level changes the vulnerability level of DVWA.
low
Submit
PHPIDS
PHPIDS v.0.6 (PHP-Intrusion Detection System) is a security
You can enable PHPIDS across this site for the duration of yo
PHPIDS is currently disabled. [enable PHPIDS]
[Simulate attack] - [View IDS log]

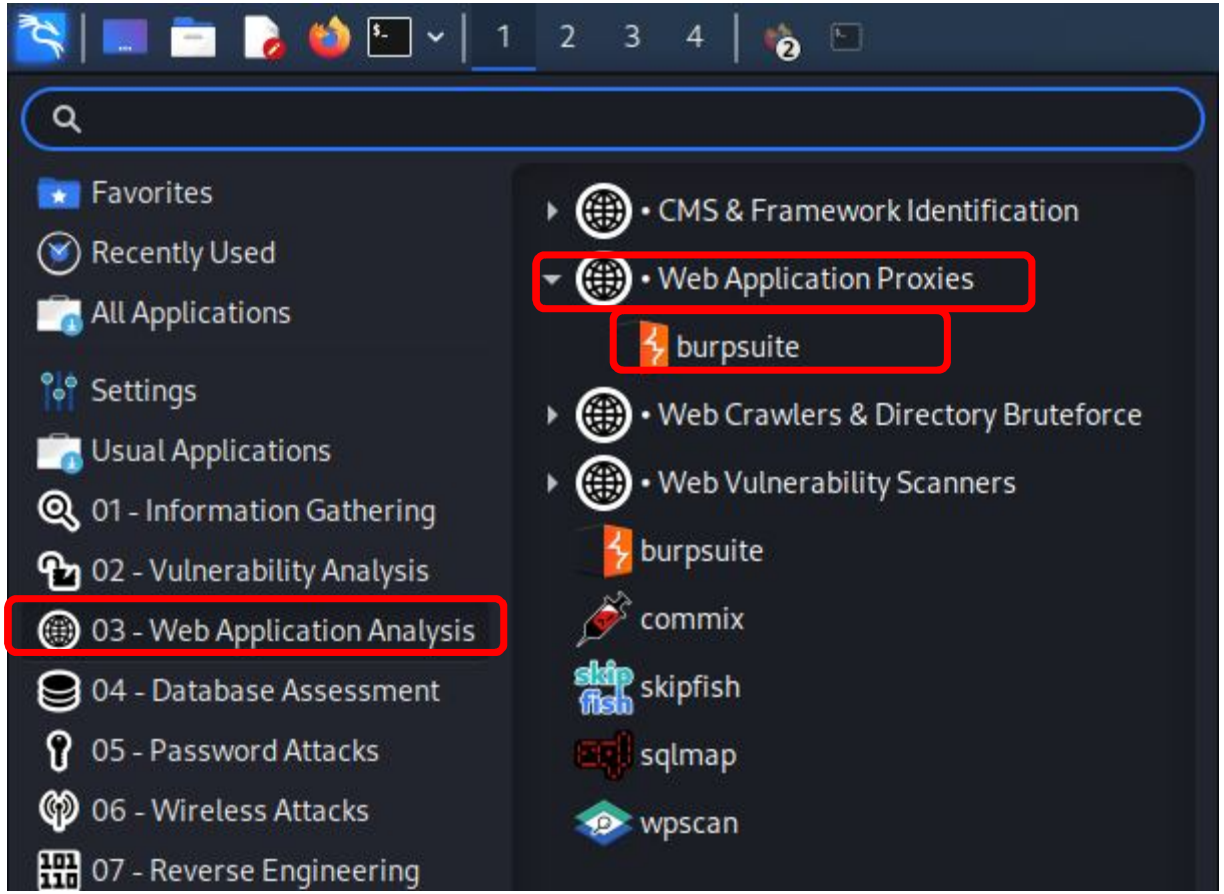
일반적인 HTTP Proxy 설정 환경

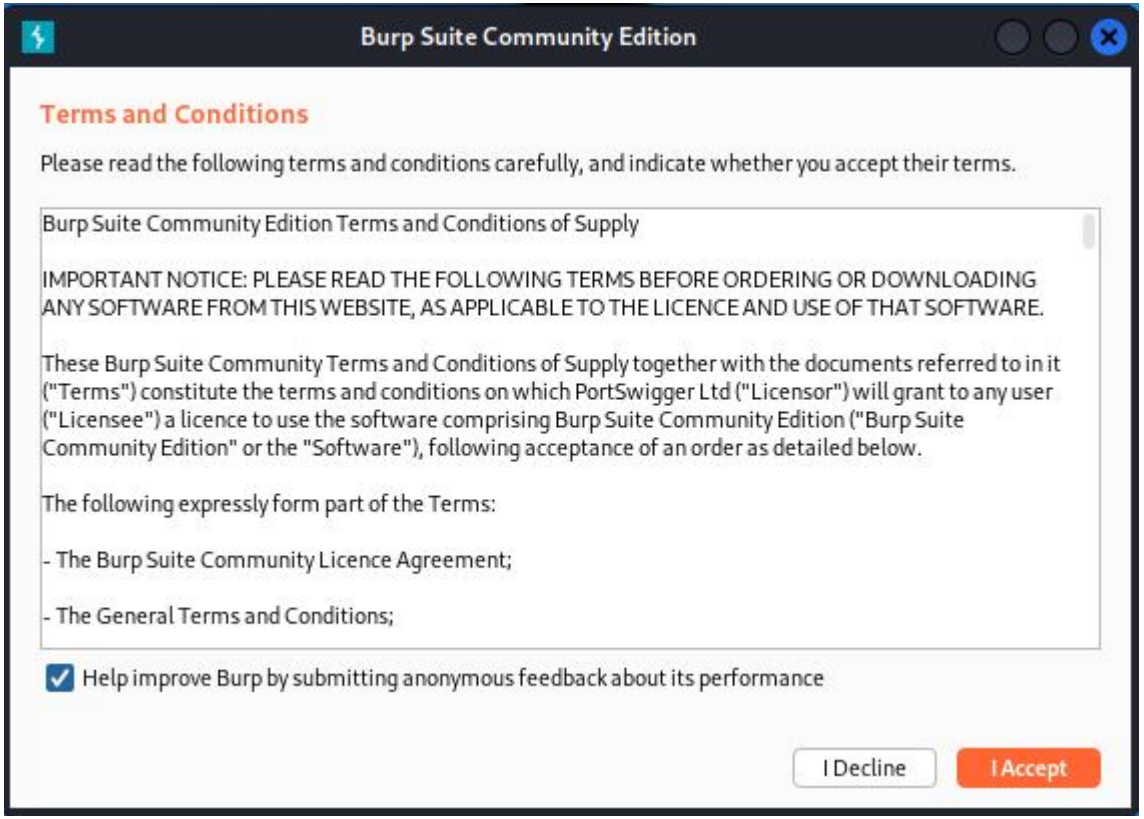


Burp Suite을 이용한 Self HTTP Proxy 설정



Burp Suit(버프 스위트) 활성화와 환경설정




Burp Suite Community Edition
Terms and Conditions
Please read the following terms and conditions carefully, and indicate whether you accept their terms.
Burp Suite Community Edition Terms and Conditions of Supply
IMPORTANT NOTICE: PLEASE READ THE FOLLOWING TERMS BEFORE ORDERING OR DOWNLOADING ANY SOFTWARE FROM THIS WEBSITE, AS APPLICABLE TO THE LICENCE AND USE OF THAT SOFTWARE.
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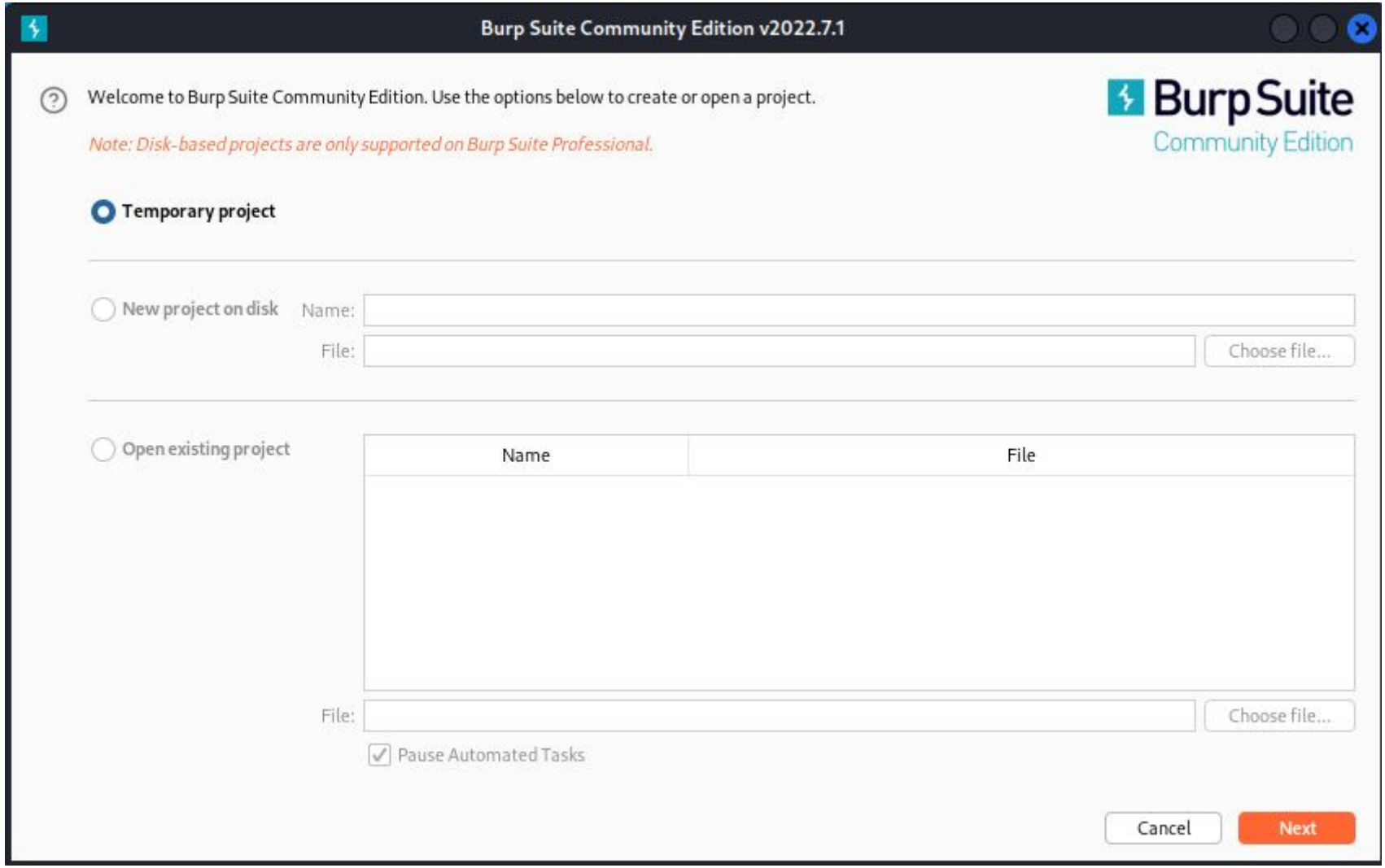
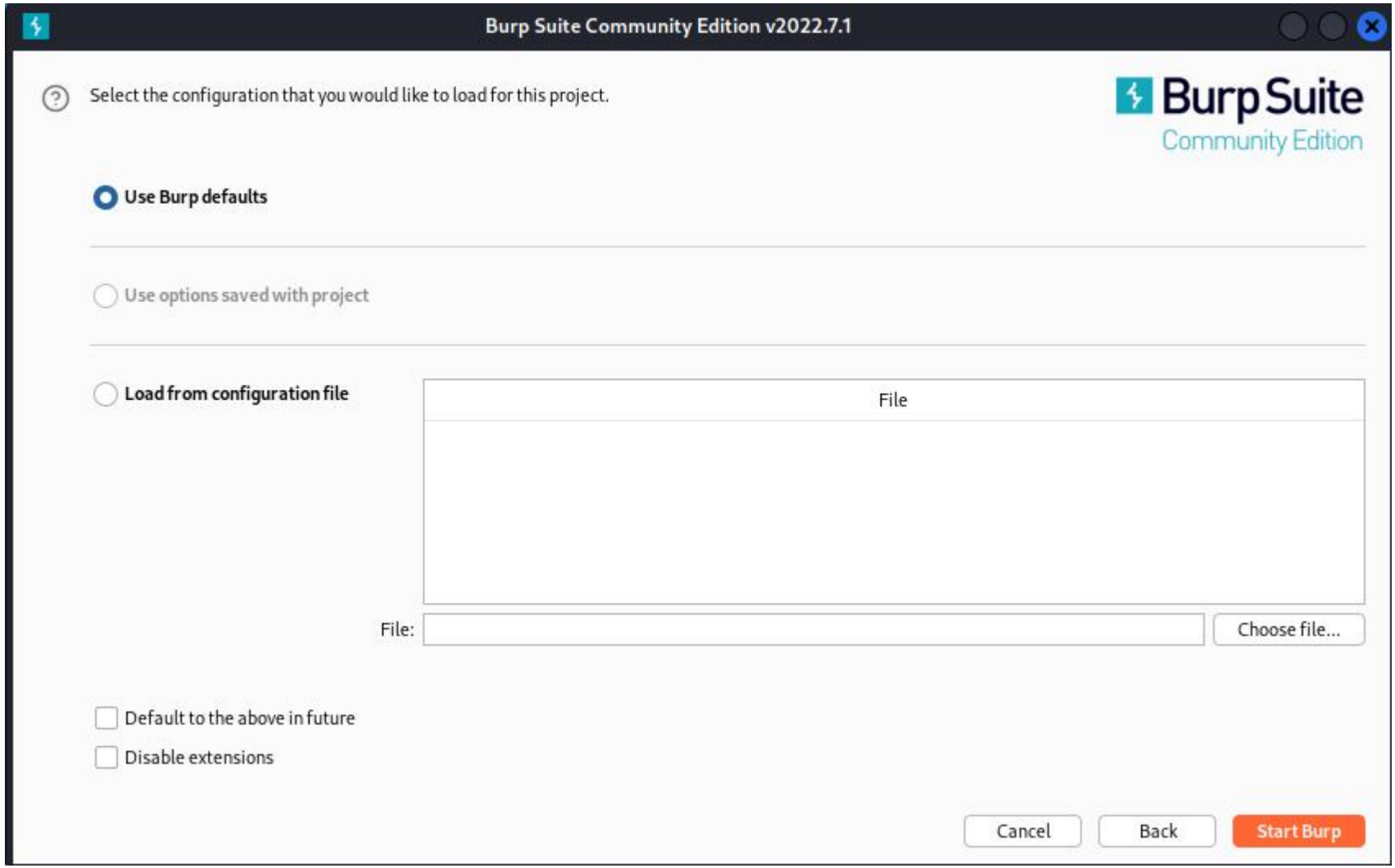
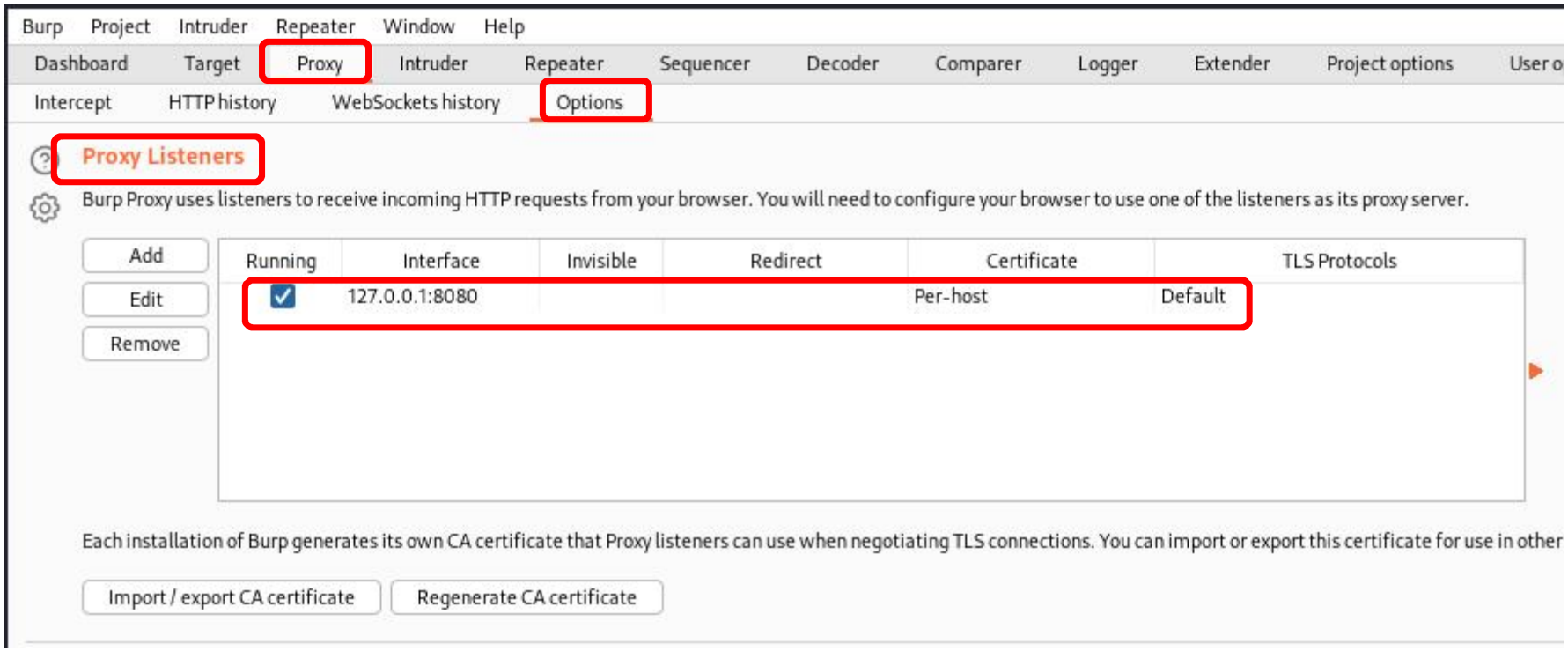
The following expressly form part of the Terms:
- The Burp Suite Community Licence Agreement;
- The General Terms and Conditions;
Help improve Burp by submitting anonymous feedback about its performance
I Decline
I Accept


Burp Suite Community Edition v2022.7.1
Welcome to Burp Suite Community Edition. Use the options below to create or open a project.
Note: Disk-based projects are only supported on Burp Suite Professional.
Burp Suite
Community Edition
Temporary project
New project on disk
Name:
File:
Choose file...
Open existing project
Name
File
File:
Choose file...
Pause Automated Tasks
Cancel
Next


Burp Suite Community Edition v2022.7.1
Select the configuration that you would like to load for this project.
Burp Suite
Community Edition
Use Burp defaults
Use options saved with project
Load from configuration file
File
File:
Choose file...
Default to the above in future
Disable extensions
Cancel
Back
Start Burp


Burp
Project
Intruder
Repeater
Window
Help
Dashboard
Target
Proxy
Intruder
Repeater
Sequencer
Decoder
Comparer
Logger
Extender
Project options
Intercept
HTTP history
WebSockets history
Options
Proxy Listeners
Burp Proxy uses listeners to receive incoming HTTP requests from your browser. You will need to configure your browser to use one of the listeners as its proxy server.
Add
Edit
Remove
Running
Interface
Invisible
Redirect
Certificate
TLS Protocols
127.0.0.1:8080
Per-host
Default
Each installation of Burp generates its own CA certificate that Proxy listeners can use when negotiating TLS connections. You can import or export this certificate for use in other
Import / export CA certificate
Regenerate CA certificate

Intercept Client Requests

Use these settings to control which requests are stalled for viewing and editing in the Intercept tab.

☑ Intercept requests based on the following rules: *Master interception is turned off*

Add | Edit | Remove | Up | Down

Enabled	Operator	Match type	Relationship	Condition
☑		File extension	Does not match	(^gif$\|^jpg$|^png$\|^css$|^js$\|^ico$|^sv...
☐	Or	Request	Contains parameters	
☐	Or	HTTP method	Does not match	(get|post)
☐	And	URL	Is in target scope	

☐ Automatically fix missing or superfluous new lines at end of request

☑ Automatically update Content-Length header when the request is edited

Intercept Server Responses

Use these settings to control which responses are stalled for viewing and editing in the Intercept tab.

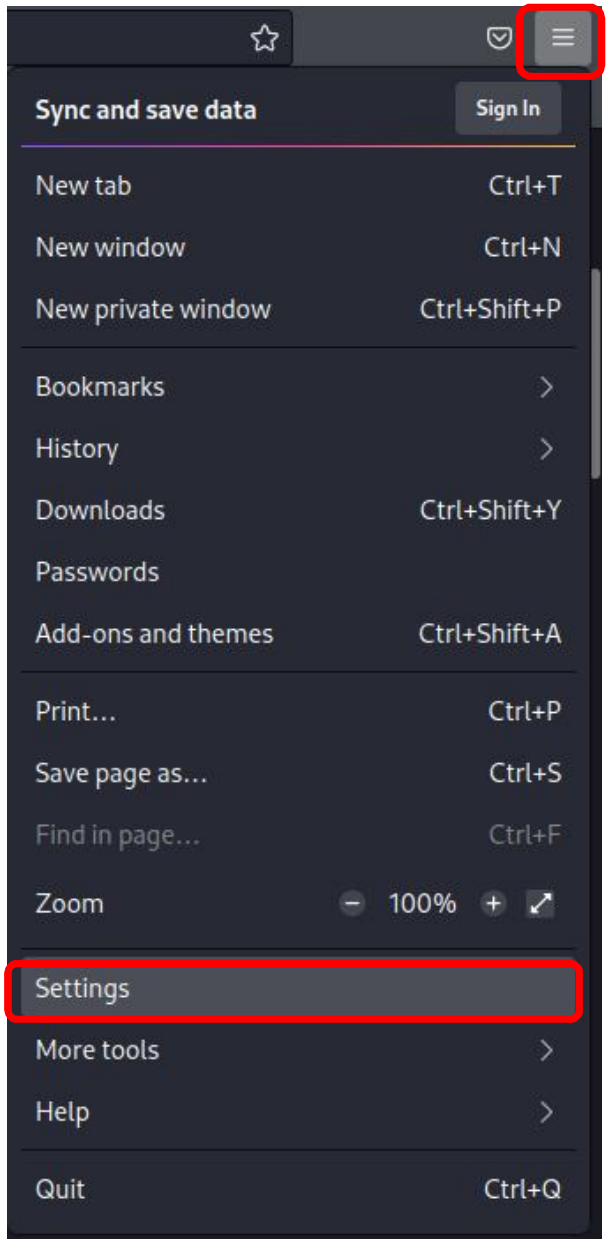
☑ Intercept responses based on the following rules: *Master interception is turned off*

Add | Edit | Remove | Up | Down

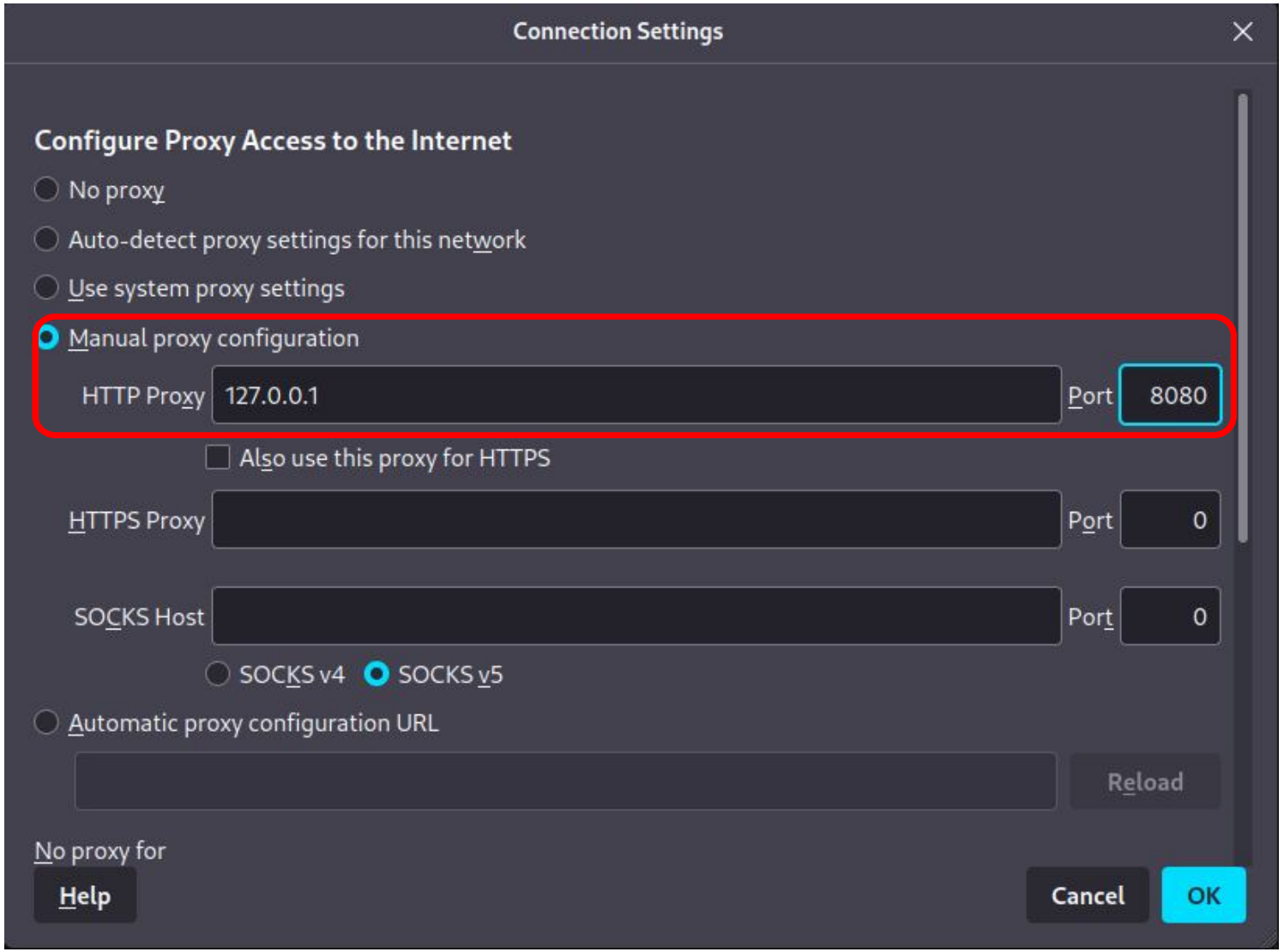
Enabled	Operator	Match type	Relationship	Condition
☑		Content type header	Matches	text
☐	Or	Request	Was modified	
☐	Or	Request	Was intercepted	
☐	And	Status code	Does not match	^304$
☐	And	URL	Is in target scope	

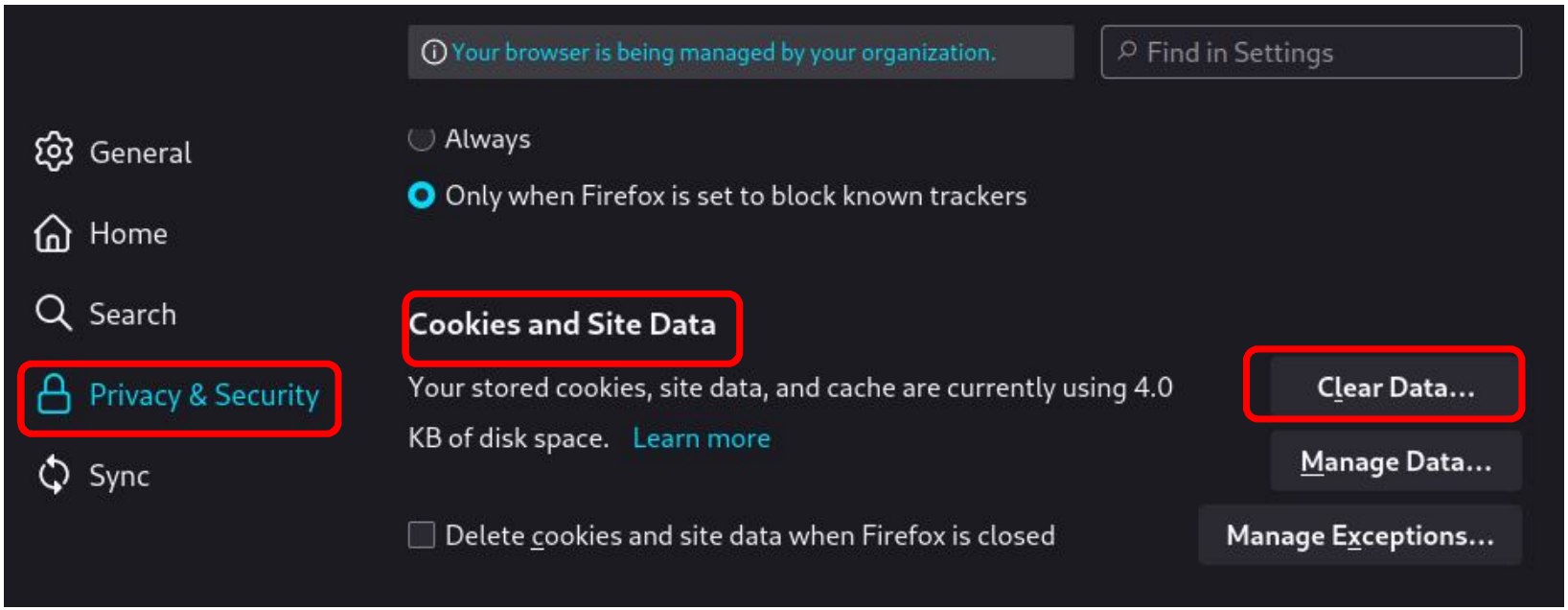
☑ Automatically update Content-Length header when the response is edited

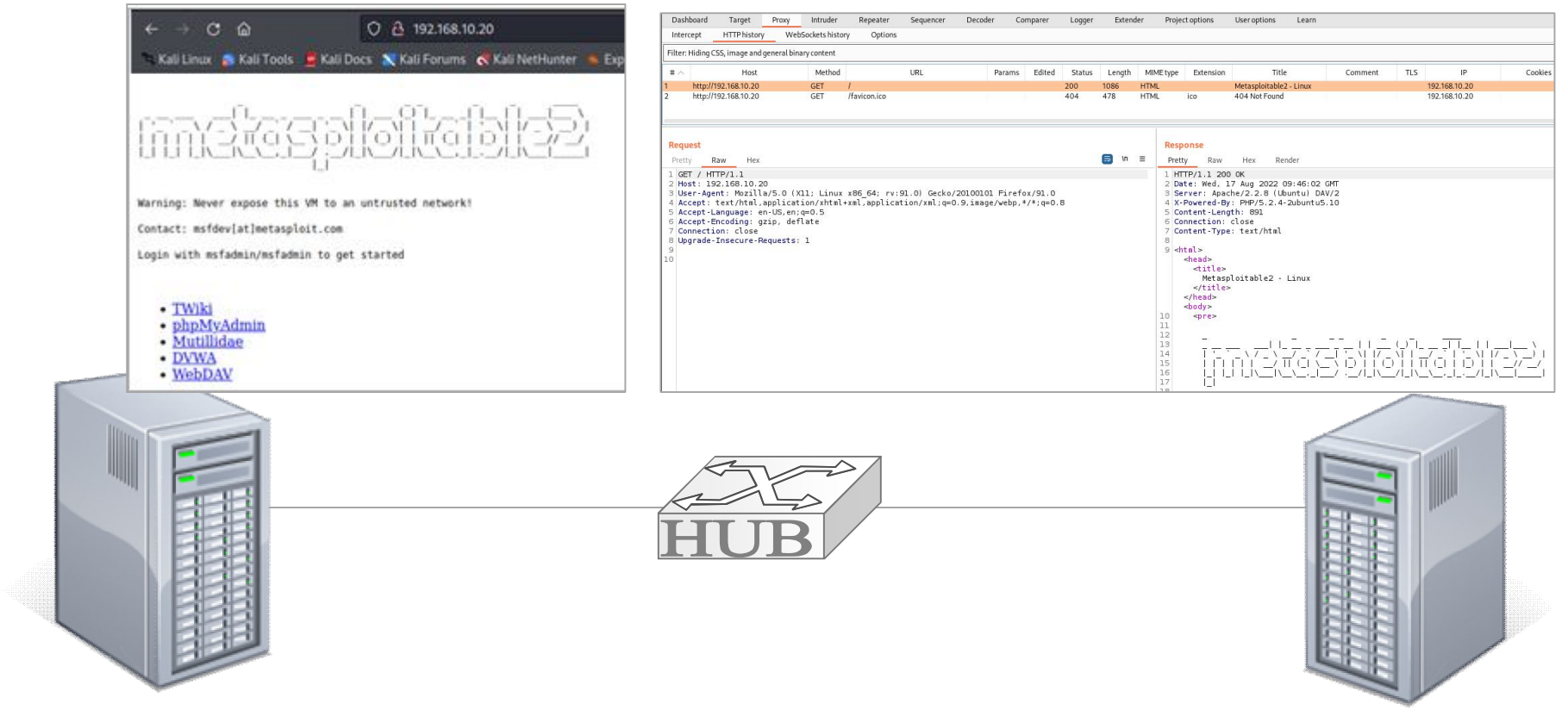
Web Browser Proxy 설정



Setting > General > Network settings > Settings




Your browser is being managed by your organization.
Find in Settings
General
Home
Search
Privacy & Security
Sync
Always
Only when Firefox is set to block known trackers
Cookies and Site Data
Your stored cookies, site data, and cache are currently using 4.0 KB of disk space. Learn more
Clear Data...
Manage Data...
Delete cookies and site data when Firefox is closed
Manage Exceptions...



Kali Attacker

Metasplitable Target